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Application of Digital Technologies to the Agricultural Learning Outcomes of Grade 9 Students in Technology and Livelihood Education: Basis for a Strategic Plan

Charisse H. Reyes
University of Perpetual Help System DALTA, Las Piñas City, Philippines
charisse.reyes@deped.gov.ph

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Corresponding Author:	Charisse H. Reyes charisse.reyes@deped.gov.ph
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Abstract

This study examined the application of digital technologies in enhancing the agricultural learning outcomes of Grade 9 students in Technology and Livelihood Education (TLE) during the School Year 2025–2026. Specifically, the study assessed the extent of digital technology utilization in terms of accessibility and usage, student engagement and participation levels, digital skills and information literacy, and practical application and quality of materials. It also determined the level of agricultural learning outcomes before and after the application of digital technology tools and examined the significant differences and relationships among the variables investigated. The study employed a descriptive-correlational research design involving 130 respondents composed of 30 TLE teachers and 100 Grade 9 students from one secondary school in the Division of Cavite. A researcher-made questionnaire-checklist served as the primary data-gathering instrument and was validated by experts prior to administration. Statistical tools such as weighted mean, paired t-test, and Pearson r correlation were utilized to analyze the gathered data. Findings revealed that digital technology tools were highly utilized in enhancing agricultural learning outcomes across all measured dimensions. Significant differences were found in the assessments of teachers and students regarding the application of digital technologies. Results further indicated that students' agricultural learning outcomes improved significantly after exposure to digital technology tools, as reflected in the posttest scores. Moreover, a significant relationship existed between the extent of digital technology application and students' agricultural learning outcomes. The findings underscore the importance of integrating digital technologies into agricultural education to improve engagement, practical skills, and academic achievement, thereby serving as a basis for the development of a strategic plan for TLE instruction.

Keywords: *digital technologies, agricultural learning outcomes, Technology and Livelihood Education, digital literacy, student engagement*

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Introduction

The integration of digital technologies into contemporary educational settings has transformed instructional delivery, learner engagement, and knowledge acquisition across multiple disciplines. In Technology and Livelihood Education (TLE), particularly within agricultural education, digital technologies have emerged as valuable tools for improving students' understanding of agricultural concepts and strengthening practical competencies. The increasing accessibility of multimedia resources, virtual simulations, online platforms, and interactive digital applications has enabled educators to create more dynamic and learner-centered instructional environments. These innovations have become especially important in technical-vocational subjects where experiential learning and practical application are essential components of effective instruction.

Agricultural education remains a significant component of the Philippine educational system because agriculture continues to play a central role in national development, food security, and rural livelihood sustainability. Despite its importance, agricultural instruction in many secondary schools continues to rely heavily on traditional teaching approaches such as textbook-based discussions and limited hands-on activities. While these conventional methods provide foundational knowledge, they may no longer fully address the learning preferences and technological orientation of present-day learners who are increasingly exposed to digital environments. Consequently, integrating digital technologies into agricultural education has become necessary to improve instructional relevance, learner engagement, and educational effectiveness.

Globally, several studies have demonstrated the educational benefits of integrating digital tools into agricultural instruction. Interactive simulations, virtual laboratories, multimedia presentations, and online collaborative platforms have been shown to improve students' comprehension of complex agricultural processes while promoting active participation and practical skill development. Digital learning tools also provide learners with immediate access to updated information, enabling them to connect theoretical agricultural concepts with real-world applications. Johnson et al. (2020) reported that interactive digital applications significantly enhanced students' motivation and retention of agricultural knowledge, suggesting that technology-enhanced learning environments contribute positively to educational outcomes.

In addition to improving conceptual understanding, digital technologies also support differentiated instruction by accommodating various learning styles and learner needs. Visual learners benefit from multimedia demonstrations and virtual simulations, while kinesthetic learners gain opportunities to engage in interactive tasks and digital problem-solving activities. These technology-enhanced instructional strategies foster deeper cognitive engagement and encourage students to participate more actively in classroom activities. Moreover, digital tools promote collaborative learning through online discussions, group projects, and shared digital platforms, thereby enhancing communication skills and peer interaction among learners. Such

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features are particularly valuable in agricultural education, where collaboration, innovation, and practical problem-solving are critical competencies.

The growing emphasis on technology integration in education became even more pronounced during the COVID-19 pandemic, which accelerated the adoption of remote and digital learning modalities worldwide. Educational institutions were compelled to transition from traditional face-to-face instruction to online and blended learning approaches to ensure continuity of education. In the Philippine context, this transition highlighted both the opportunities and challenges associated with digital learning implementation. While some schools successfully integrated digital technologies into instruction, many institutions encountered difficulties related to internet connectivity, device availability, digital literacy, and technological infrastructure. These challenges were particularly evident in rural and underserved areas where access to digital resources remained limited.

Despite the increasing adoption of digital technologies in education, localized empirical research examining their actual impact on agricultural learning outcomes among secondary school students remains limited. Existing studies often focus broadly on digital integration without specifically investigating how digital technologies influence students' comprehension, practical skills, participation, and academic achievement within the context of Technology and Livelihood Education. Furthermore, many international studies are situated in developed countries with greater technological readiness, making it difficult to generalize findings directly to the Philippine educational environment where infrastructural and socio-economic conditions differ substantially.

Another critical issue associated with digital technology integration is the disparity in accessibility and digital competency among learners and educators. Unequal access to technological devices, unstable internet connectivity, and varying levels of digital literacy may significantly affect students' ability to maximize the benefits of technology-enhanced instruction. Teachers may also encounter challenges in effectively integrating digital resources due to insufficient training and limited institutional support. These constraints may hinder the successful implementation of digital learning initiatives and potentially widen existing educational inequalities. Therefore, evaluating the actual effectiveness of digital technologies in improving agricultural learning outcomes is essential for developing responsive instructional strategies and educational policies.

The present study was anchored on Constructivist Learning Theory, which emphasizes that learners actively construct knowledge through interaction, exploration, and experience. In the context of digital agricultural education, constructivism supports the use of interactive technologies that encourage active participation, problem-solving, and experiential learning. Digital tools such as simulations, multimedia resources, and virtual learning environments enable students to engage with agricultural concepts more meaningfully by allowing them to explore realistic scenarios and apply acquired knowledge in simulated contexts. Through these

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interactive experiences, learners are able to develop deeper understanding, critical thinking skills, and practical competencies essential in agricultural education.

This study specifically examined the extent of application of digital technology tools in enhancing the agricultural learning outcomes of Grade 9 students in Technology and Livelihood Education in terms of accessibility and usage, student engagement and participation levels, digital skills and information literacy, and practical application and quality of materials. It likewise determined the level of students' agricultural learning outcomes before and after the application of digital technology tools as reflected in their pretest and posttest scores. Additionally, the study investigated whether significant differences and relationships existed among the identified variables.

The significance of this investigation lies in its potential contribution to improving instructional practices, curriculum development, and strategic educational planning in agricultural education. By identifying how digital technologies influence students' learning outcomes, the study provides evidence-based insights that may guide educators, school administrators, and policymakers in designing more responsive and technology-enhanced agricultural learning environments. The findings may also contribute to strengthening digital literacy and practical agricultural competencies among learners, preparing them for future academic, vocational, and livelihood opportunities in an increasingly technology-driven agricultural sector.

Methodology

This study employed a descriptive-correlational research design to examine the application of digital technologies in enhancing the agricultural learning outcomes of Grade 9 students in Technology and Livelihood Education (TLE). The descriptive component of the study enabled the researcher to systematically assess the extent to which digital technology tools were utilized in agricultural instruction, particularly in terms of accessibility and usage, student engagement and participation levels, digital skills and information literacy, and practical application and quality of materials. The correlational aspect of the design was utilized to determine the significant relationship between the extent of application of digital technology tools and the agricultural learning outcomes of students as reflected in their posttest scores. This design was considered appropriate because it allowed the researcher to describe existing conditions and examine relationships among variables without manipulating the study environment.

The study was conducted in one secondary school in the Division of Cavite during the School Year 2025–2026. The respondents consisted of teachers teaching Technology and Livelihood Education and Grade 9 students enrolled in agricultural learning subjects. The general population of the study included all TLE teachers and students within the identified institution. Using probability sampling through total enumeration, all qualified participants

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were included in the study to ensure comprehensive representation of the target population. The final sample consisted of 30 teachers and 100 students, yielding a total of 130 respondents. These respondents evaluated the extent of application of digital technology tools in enhancing agricultural learning outcomes across the identified dimensions of the study.

A researcher-made questionnaire-checklist served as the primary data-gathering instrument. The instrument was specifically designed to gather information relevant to the objectives of the study and was divided into several parts. The first part elicited the demographic profile of the respondents. The second part assessed the extent of application of digital technology tools in enhancing agricultural learning outcomes in terms of accessibility and usage, student engagement and participation levels, digital skills and information literacy, and practical application and quality of materials. The third part determined the level of agricultural learning outcomes before and after the application of digital technology tools through pretest and posttest measures.

To ensure consistency in responses and facilitate quantitative analysis, the questionnaire utilized a four-point Likert scale with corresponding verbal interpretations. Responses ranging from 3.26 to 4.00 were interpreted as “Highly Evident,” values from 2.51 to 3.25 as “Evident,” values from 1.76 to 2.50 as “Slightly Evident,” and values from 1.00 to 1.75 as “Not at All.” This scaling procedure enabled the researcher to systematically quantify respondents’ perceptions regarding the application of digital technologies in agricultural education.

Prior to the actual conduct of the study, the researcher subjected the instrument to validation and reliability testing procedures to establish its content validity and internal consistency. The questionnaire was initially reviewed by the research adviser to obtain preliminary comments and recommendations for improvement. After revisions were made, the instrument was further evaluated by experts in educational research, test construction, and the field of Technology and Livelihood Education. Validators included professors from the University of Perpetual Help System DALTA and teachers from one secondary school in the Division of Cavite who possessed relevant expertise and professional experience. Their recommendations and suggestions were incorporated into the final version of the instrument to strengthen its validity and clarity.

A pilot test was subsequently conducted to determine the reliability of the instrument. Using the Statistical Package for the Social Sciences (SPSS), Cronbach’s Alpha was computed to assess the internal consistency of the questionnaire items. The reliability testing procedure ensured that the instrument consistently measured the constructs identified in the study. The researcher carefully considered all expert comments and statistical findings before finalizing the instrument for administration.

The data-gathering process commenced after obtaining approval from the school administration of the selected secondary school in the Division of Cavite. Formal letters requesting permission to conduct the study were distributed to school authorities and prospective respondents. Upon approval, informed consent was secured from all participants.

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For minor student participants, parental consent was likewise obtained to ensure compliance with ethical standards involving human participants. The researcher personally explained the objectives of the study, the procedures involved, and the instructions for answering the questionnaire to ensure respondents clearly understood the nature and purpose of the investigation.

Following the orientation, the survey questionnaires were administered to the respondents, who were given adequate time to complete the instrument. After retrieval, the accomplished questionnaires were carefully checked, organized, tallied, and encoded for statistical processing. The collected data were analyzed using appropriate statistical tools through SPSS. The results obtained from the statistical analyses served as the basis for interpreting the findings and developing a strategic plan intended to further improve the application of digital technology tools in enhancing agricultural learning outcomes in Technology and Livelihood Education.

Several statistical tools were employed to analyze and interpret the gathered data. Weighted mean was utilized to determine the extent of application of digital technology tools in terms of accessibility and usage, student engagement and participation levels, digital skills and information literacy, and practical application and quality of materials. Paired t-test was used to determine significant differences in the assessments of the two groups of respondents and to examine differences in students' agricultural learning outcomes before and after the application of digital technology tools. Pearson product-moment correlation coefficient (Pearson r) was employed to determine the significant relationship between the extent of application of digital technology tools and students' posttest scores in agricultural learning outcomes.

Ethical considerations were strictly observed throughout the conduct of the study to protect the rights, welfare, and confidentiality of all participants. The researcher ensured voluntary participation by informing respondents that they could withdraw from the study at any stage without penalty or negative consequences. Confidentiality and anonymity were maintained by removing identifying information from all collected data and securely storing the accomplished questionnaires and encoded datasets. The researcher further assured participants that their responses would be used solely for academic purposes and would not affect their academic standing or professional evaluations.

The researcher also upheld principles of honesty, transparency, and objectivity in the collection, analysis, and interpretation of data. No data manipulation, selective reporting, or misrepresentation of findings was committed during the conduct of the investigation. Respect for participants' dignity, cultural background, and emotional well-being was likewise maintained throughout the research process. At the conclusion of the study, findings were communicated appropriately to contribute to the improvement of instructional practices and digital technology integration within agricultural education.

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Results

This section presents the findings of the study concerning the application of digital technology tools in enhancing the agricultural learning outcomes of Grade 9 students in Technology and Livelihood Education (TLE). The presentation of data follows the sequence of the research problems and focuses on the extent of application of digital technologies, differences in respondents' assessments, students' pretest and posttest learning outcomes, and the relationship between digital technology application and agricultural learning outcomes. The findings are presented objectively using statistically significant results supported by concise analytical narration.

The first objective of the study determined the extent of application of digital technology tools in enhancing the agricultural learning outcomes of Grade 9 students in Technology and Livelihood Education in terms of accessibility and usage, student engagement and participation levels, digital skills and information literacy, and practical application and quality of materials.

Extent of Application of Digital Technology Tools

The findings revealed that both teachers and students assessed the application of digital technology tools in agricultural learning as highly evident across all dimensions measured in the study. Results showed consistently high composite means, indicating that digital tools were frequently integrated into instructional activities and learning experiences in agricultural education.

Table 1

Summary of the Respondents' Assessment on the Extent of Application of Digital Technology Tools in Enhancing Agricultural Learning Outcomes

Indicators	Teachers Mean	VI	Students Mean	VI
Accessibility and Usage	3.91	Highly Evident	3.57	Highly Evident
Student Engagement and Participation Levels	3.92	Highly Evident	3.55	Highly Evident
Digital Skills and Information Literacy	3.90	Highly Evident	3.51	Highly Evident
Practical Application and Quality of Materials	3.93	Highly Evident	3.56	Highly Evident
Overall Mean	**3.91**	**Highly Evident**	**3.55**	**Highly Evident**

In terms of accessibility and usage, teachers obtained a composite mean of 3.91 while students recorded a composite mean of 3.57, both verbally interpreted as highly evident. The

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findings indicate that digital technology tools were consistently accessible and actively utilized in agricultural instruction. Teachers regularly integrated digital resources into classroom activities, ensured students' access to technology tools, and incorporated updated digital materials to support agricultural learning. Students likewise perceived that digital resources were readily available and effectively used to facilitate understanding of agricultural concepts and practices.

Regarding student engagement and participation levels, teachers registered a composite mean of 3.92 while students obtained a composite mean of 3.55, both interpreted as highly evident. The findings suggest that digital technologies significantly enhanced students' participation, motivation, and interaction during agricultural lessons. Teachers frequently utilized interactive digital tools, multimedia resources, and collaborative online activities that encouraged active student involvement in agricultural learning tasks and projects. Students demonstrated heightened engagement when digital resources were integrated into classroom instruction.

For digital skills and information literacy, teachers achieved a composite mean of 3.90 while students obtained a composite mean of 3.51, both verbally interpreted as highly evident. Results indicate that digital technology integration contributed positively to the development of students' digital competencies and information literacy skills. Respondents observed that students were guided in evaluating online agricultural information, utilizing digital tools for research and problem-solving, and practicing responsible and ethical use of technology in agricultural studies. The findings demonstrate that digital technologies supported the acquisition of essential 21st-century skills necessary for modern agricultural learning environments.

Similarly, in terms of practical application and quality of materials, teachers obtained a composite mean of 3.93 while students recorded a composite mean of 3.56, both verbally interpreted as highly evident. The results indicate that digital technology tools effectively facilitated practical agricultural learning experiences and enhanced the quality of instructional materials used in TLE classes. Teachers incorporated high-quality and updated digital resources that supported students' understanding of agricultural processes, practical applications, and skill development. The findings further suggest that digital tools strengthened experiential learning by connecting theoretical agricultural concepts to practical activities and real-world contexts.

Overall, the results demonstrate that the application of digital technology tools was consistently and extensively integrated into agricultural learning activities for Grade 9 students. Both teachers and students recognized the substantial role of digital technologies in enhancing accessibility, engagement, digital literacy, and practical agricultural learning experiences. These findings suggest that digital technologies have become valuable instructional resources in improving the quality of agricultural education within Technology and Livelihood Education programs.

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Difference in the Assessment of the Two Groups of Respondents

The second objective of the study determined whether a significant difference existed in the assessments of teachers and students regarding the extent of application of digital technology tools in enhancing agricultural learning outcomes.

Table 2

Difference in the Assessment of the Two Groups of Respondents on the Extent of Application of Digital Technology Tools

Indicators	t-value	p-value	Decision	Interpretation
Accessibility and Usage	3.116	.004	Reject Ho	Significant
Student Engagement and Participation Levels	3.334	.002	Reject Ho	Significant
Digital Skills and Information Literacy	5.702	.000	Reject Ho	Significant
Practical Application and Quality of Materials	4.888	.000	Reject Ho	Significant
Overall	**5.948**	**.000**	**Reject Ho**	**Significant**

The findings revealed statistically significant differences between the assessments of teachers and students across all measured indicators. Accessibility and usage obtained a p-value of .004, student engagement and participation levels registered .002, digital skills and information literacy yielded .000, and practical application and quality of materials also obtained .000, all of which were below the .05 level of significance. The overall p-value of .000 confirmed the rejection of the null hypothesis. These findings indicate that teachers and students differed significantly in their perceptions regarding the extent of application of digital technology tools in agricultural education.

The differences in perceptions may be attributed to variations in experiences, levels of interaction with digital technologies, and expectations regarding technology integration in instruction. Teachers, as implementers of instructional strategies, may possess broader perspectives concerning the pedagogical use and effectiveness of digital tools, whereas students primarily evaluate technology integration based on usability, engagement, and learning experiences. Despite these differences, both groups consistently recognized the high utilization of digital technologies in agricultural education.

Level of Agricultural Learning Outcomes Before and After the Application of Digital Technology Tools

The third objective of the study examined the level of agricultural learning outcomes of Grade 9 students before and after the application of digital technology tools as reflected in their pretest and posttest scores.

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Table 3

Level of Agricultural Learning Outcomes Before and After the Application of Digital Technology Tools

Assessment	Mean Score	Interpretation
Pretest	Lower Performance Level	Before Application
Posttest	Improved Performance Level	After Application

The findings indicated that students' agricultural learning outcomes improved after the application of digital technology tools in TLE instruction. Posttest scores demonstrated better performance compared to pretest scores, suggesting that digital technologies positively contributed to students' acquisition of agricultural knowledge and skills. The improvement in performance reflects the effectiveness of digital instructional resources in supporting students' understanding of agricultural concepts, participation in learning activities, and practical skill development.

Significant Difference Between Pretest and Posttest Scores

The fourth objective of the study determined whether a significant difference existed between the pretest and posttest scores of Grade 9 students following the application of digital technology tools.

The findings revealed a statistically significant difference between students' pretest and posttest scores, indicating that exposure to digital technology tools significantly enhanced agricultural learning outcomes. The rejection of the null hypothesis confirms that the application of digital technologies contributed positively to students' academic performance in Technology and Livelihood Education. The observed improvement suggests that digital tools facilitated more effective instructional delivery, increased learner engagement, and strengthened students' comprehension and practical application of agricultural concepts.

Relationship Between Application of Digital Technology Tools and Agricultural Learning Outcomes

The final objective of the study examined whether a significant relationship existed between the extent of application of digital technology tools and the agricultural learning outcomes of Grade 9 students as reflected in posttest scores.

Table 4

Relationship Between Application of Digital Technology Tools and Agricultural Learning Outcomes

Variables	Relationship		Interpretation
Application of Digital Technology Tools and Posttest Scores	Significant Relationship	Positive	Significant

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The results revealed a significant positive relationship between the extent of application of digital technology tools and students' agricultural learning outcomes. This finding indicates that greater utilization of digital technologies corresponded with improved posttest performance among Grade 9 students. The relationship suggests that digital technologies effectively supported agricultural instruction by enhancing accessibility to learning resources, strengthening engagement, improving digital literacy, and facilitating practical application of knowledge. The findings further imply that the integration of digital tools can contribute meaningfully to improving educational quality and student achievement in Technology and Livelihood Education.

Discussion

The findings of the study demonstrate that the application of digital technology tools significantly contributed to enhancing the agricultural learning outcomes of Grade 9 students in Technology and Livelihood Education (TLE). The consistently high assessments from both teachers and students across all dimensions of digital technology application suggest that digital integration has become an essential component of agricultural instruction. These findings support the growing body of literature emphasizing the transformative role of digital technologies in promoting learner engagement, improving instructional quality, and strengthening practical competencies in technical-vocational education.

The high level of accessibility and usage of digital technology tools indicates that learners were provided with meaningful opportunities to engage with technology-enhanced agricultural instruction. The findings suggest that teachers effectively integrated digital platforms, multimedia resources, and updated technological tools into classroom activities, thereby creating a more supportive and interactive learning environment. This result aligns with Constructivist Learning Theory, which posits that learners construct knowledge more effectively when they actively interact with instructional materials and learning environments. Through digital technologies, students were able to access information, manipulate learning resources, and participate in experiential activities that promoted deeper understanding of agricultural concepts.

The findings further corroborate previous studies emphasizing the importance of accessibility in maximizing the educational benefits of technology integration. Cruz (2023) reported that regular access to interactive digital platforms significantly improved students' practical skill acquisition in vocational education. Similar patterns emerged in the present study, where respondents observed that the availability and frequent use of digital tools contributed positively to agricultural learning experiences. Accessibility to technology resources appears to facilitate continuity of learning, increase instructional flexibility, and support students' independent exploration of agricultural concepts and practices.

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The results concerning student engagement and participation levels indicate that digital technologies created more interactive and motivating learning experiences for Grade 9 students. Students demonstrated heightened participation and involvement during agricultural lessons when digital resources such as multimedia presentations, interactive activities, and collaborative online tools were incorporated into instruction. These findings support the argument that digital learning environments foster active participation by transforming learners from passive recipients of information into active participants in the learning process. Engagement is particularly important in agricultural education because practical and experiential learning requires sustained learner interaction, curiosity, and collaboration.

The findings are also consistent with Mendoza's (2023) study, which found that interactive digital learning platforms significantly increased student participation and motivation in vocational education. Similarly, Thompson et al. (2023) emphasized that interactive digital features such as collaborative activities and multimedia applications improved classroom engagement and reduced learner disengagement in virtual and blended environments. These findings collectively reinforce the notion that digital technologies enhance educational effectiveness not only by improving content delivery but also by promoting meaningful learner participation and motivation.

In terms of digital skills and information literacy, the study revealed that students developed stronger competencies in accessing, evaluating, and utilizing digital agricultural information through the integration of digital technologies. This finding reflects the increasing importance of digital literacy as a fundamental educational competency in the 21st century. Agricultural education now extends beyond traditional farming concepts and increasingly incorporates digital platforms, online research, precision agriculture technologies, and information management systems. Consequently, students must acquire the ability to critically evaluate digital information, ethically use technological resources, and apply digital tools to solve agricultural problems effectively.

The findings parallel those of Garcia (2023), who reported that students consistently exposed to digital learning platforms demonstrated improved information literacy and critical analysis skills. Likewise, Johnson and Alvarez (2021) emphasized that structured digital literacy interventions significantly enhanced students' academic performance and research competencies. The present study supports these conclusions by demonstrating that digital technology integration contributes not only to subject-specific agricultural learning but also to the development of broader technological competencies necessary for lifelong learning and workforce readiness.

The high assessments related to practical application and quality of materials further highlight the effectiveness of digital technologies in supporting experiential agricultural learning. Respondents observed that digital instructional materials were updated, relevant, and effective in facilitating practical understanding of agricultural concepts and techniques. The integration of digital tools allowed students to connect theoretical knowledge with real-world agricultural applications, thereby strengthening comprehension and skill retention. This finding is significant because agricultural education heavily relies on practical experiences, simulations, demonstrations, and contextualized learning activities to develop technical competencies.

The findings align with Santos (2023), who found that interactive digital instructional materials significantly improved vocational students' practical skills and understanding of technical concepts. Similarly, Silva et al. (2022) emphasized that integrated digital and physical learning materials enhanced students' applied learning performance by combining interactive visual content with hands-on experiences. The present study therefore reinforces the importance of utilizing high-quality digital instructional materials that are contextually relevant, interactive, and aligned with practical agricultural learning objectives.

A significant finding of the study was the statistically significant difference between the assessments of teachers and students regarding the application of digital technology tools. Although both groups generally rated digital technology integration positively, variations in perceptions were evident across all dimensions. These differences may stem from disparities in technological familiarity, instructional expectations, and levels of direct engagement with digital resources. Teachers, as facilitators of instruction, may evaluate digital technologies based on pedagogical effectiveness and implementation challenges, whereas students may focus more on usability, accessibility, and engagement.

The existence of perceptual differences highlights the importance of continuous communication and collaboration between teachers and students regarding technology integration practices. It also underscores the need for professional development programs that equip teachers with advanced digital pedagogical competencies while simultaneously ensuring that students possess the necessary technological support and digital literacy skills to maximize learning opportunities. Addressing these differences is essential to achieving more equitable and effective digital learning experiences in agricultural education.

Another major finding of the study was the significant improvement in students' agricultural learning outcomes following the application of digital technology tools. The observed increase in posttest performance indicates that digital technologies positively influenced students' academic achievement and understanding of agricultural concepts. This improvement suggests that technology-enhanced instruction facilitated more engaging, interactive, and effective learning experiences that supported knowledge acquisition and practical skill development.

The significant difference between pretest and posttest scores further supports the effectiveness of digital technologies in improving learning outcomes. These findings are consistent with studies conducted by Kim and Lee (2022) and Santos et al. (2024), which reported that digital and blended learning approaches significantly enhanced students' agricultural knowledge, engagement, and practical competencies. The current study therefore contributes localized empirical evidence demonstrating that digital technologies can effectively

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improve agricultural learning outcomes among secondary school students in the Philippine educational context.

The significant positive relationship between the extent of digital technology application and students' posttest scores further strengthens the argument that technology integration contributes meaningfully to educational improvement. The findings suggest that greater utilization of digital tools corresponded with better agricultural learning outcomes, indicating that technology-enhanced instruction positively influenced student performance. This relationship highlights the importance of sustained digital integration efforts in agricultural education and supports the growing recognition of technology as a critical instructional resource in technical-vocational learning environments.

The study carries several implications for educational practice, policy, and curriculum development. Schools should continue investing in digital infrastructure, technological resources, and teacher training programs to strengthen the integration of digital technologies in agricultural education. Educational leaders and policymakers should likewise develop strategic initiatives that promote equitable access to digital resources, particularly in underserved schools and communities where technological disparities remain evident. Strengthening digital literacy among both teachers and students is also essential to ensuring effective and sustainable technology integration.

From a pedagogical perspective, the findings emphasize the value of learner-centered, technology-supported instructional approaches in improving agricultural learning outcomes. Digital technologies provide opportunities for experiential, collaborative, and inquiry-based learning experiences that align with contemporary educational goals and industry demands. By integrating interactive and contextually relevant digital resources, educators can create more engaging agricultural learning environments that prepare students for future academic, vocational, and agricultural careers.

Practical Output

Strategic Plan for Enhancing Agricultural Learning Outcomes Through Digital Technology Integration in Technology and Livelihood Education

The findings of the study demonstrated that digital technology tools significantly enhanced the agricultural learning outcomes of Grade 9 students in Technology and Livelihood Education. Results revealed highly evident utilization of digital technologies in terms of accessibility and usage, student engagement and participation, digital skills and information literacy, and practical application and quality of materials. Moreover, significant improvements in students' posttest performance and the identified positive relationship between digital technology application and agricultural learning outcomes provided empirical support for the development of a strategic instructional plan aimed at strengthening technology integration in agricultural education.

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The proposed strategic plan is designed to improve instructional delivery, learner engagement, practical skill acquisition, and digital literacy competencies among Grade 9 students in agricultural education. The plan also aims to address observed differences in teachers' and students' perceptions regarding digital technology utilization by promoting equitable access, teacher capacity-building, and learner-centered digital instructional practices. The strategic plan focuses on four major areas: digital accessibility and infrastructure enhancement, teacher professional development, student digital engagement and literacy enhancement, and instructional material improvement and monitoring.

The first component of the strategic plan focuses on strengthening digital accessibility and infrastructure. Findings indicated that accessibility and usage of digital technology tools were highly evident; however, variations in respondents' perceptions suggest that some students may still experience challenges related to equitable access and utilization of technological resources. To address this concern, the school may strengthen its digital infrastructure by improving internet connectivity, increasing the availability of digital devices, and ensuring that classrooms are equipped with functional multimedia resources and technology-supported instructional facilities. Establishing accessible digital learning spaces and providing technical support services may further ensure continuous and effective utilization of digital resources in agricultural instruction.

The second component of the strategic plan emphasizes continuous professional development for teachers. Since teachers serve as primary facilitators of technology integration, enhancing their digital pedagogical competencies is essential to sustaining effective instructional practices. The study findings demonstrated that digital technologies significantly improved student engagement, participation, and learning outcomes, highlighting the importance of competent instructional implementation. Regular seminars, workshops, and training programs may therefore be conducted to strengthen teachers' knowledge of digital instructional strategies, multimedia integration, virtual simulations, online assessment tools, and technology-based agricultural learning applications. Collaborative learning communities among teachers may likewise be established to encourage sharing of best practices, instructional innovations, and effective technology integration approaches.

The third component centers on enhancing student engagement, participation, and digital literacy skills. Results revealed that digital technologies significantly improved students' involvement in agricultural learning activities and strengthened their information literacy competencies. To further maximize these benefits, teachers may design more interactive, collaborative, and inquiry-based learning activities using digital platforms and multimedia resources. Students may be encouraged to participate in technology-supported agricultural projects, virtual demonstrations, online collaborative tasks, and digital research activities that promote critical thinking, problem-solving, and independent learning. Additionally, digital citizenship and information literacy sessions may be incorporated into instructional activities to strengthen students' responsible and ethical use of digital resources.

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The fourth component focuses on improving the quality, relevance, and practical application of digital instructional materials. Findings showed that practical application and quality of materials were highly evident and contributed positively to agricultural learning outcomes. To sustain these gains, teachers and curriculum planners may continuously update and contextualize digital instructional materials to ensure alignment with current agricultural practices, industry trends, and learner needs. Instructional materials may include interactive multimedia lessons, agricultural simulation software, instructional videos, digital manuals, and virtual field demonstrations that allow students to apply theoretical knowledge in practical contexts. Emphasis should also be placed on integrating localized and sustainable agricultural content that reflects real-life agricultural scenarios relevant to the Philippine context.

To ensure effective implementation of the strategic plan, systematic monitoring and evaluation procedures may be conducted periodically. School administrators and instructional leaders may assess the effectiveness of digital technology integration through classroom observations, student performance monitoring, teacher feedback, and learner satisfaction surveys. Evaluation findings may serve as bases for refining instructional strategies, addressing implementation challenges, and sustaining continuous improvement initiatives in agricultural education. Monitoring mechanisms may also help determine whether digital technologies continue to positively influence student engagement, digital literacy, and agricultural learning outcomes over time.

The proposed strategic plan serves as a responsive and evidence-based framework for strengthening digital technology integration in Technology and Livelihood Education. Grounded in the findings of the study, the plan supports the development of more engaging, inclusive, and technology-responsive agricultural learning environments capable of improving students' academic achievement, practical competencies, and readiness for future agricultural and vocational opportunities.

Conclusion

The study established that the application of digital technology tools significantly enhanced the agricultural learning outcomes of Grade 9 students in Technology and Livelihood Education. Findings revealed that digital technologies were highly evident in terms of accessibility and usage, student engagement and participation levels, digital skills and information literacy, and practical application and quality of materials. Both teachers and students recognized the substantial contribution of digital tools in improving instructional delivery and facilitating more interactive, meaningful, and learner-centered agricultural education experiences. The consistent integration of digital technologies into classroom instruction demonstrated their value as effective educational resources capable of supporting knowledge acquisition, skill development, and learner motivation.

The study further revealed significant differences in the assessments of teachers and students regarding the extent of digital technology application in agricultural learning. These

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differences suggest that although digital technologies were positively perceived by both groups, variations existed in their experiences, expectations, and levels of interaction with technology-enhanced instruction. This finding underscores the importance of strengthening communication, instructional alignment, and support systems to ensure that both educators and learners maximize the benefits of digital integration in Technology and Livelihood Education.

Results also confirmed that the agricultural learning outcomes of Grade 9 students improved significantly after exposure to digital technology tools, as evidenced by differences between pretest and posttest performance. The improvement in students' academic achievement indicates that digital technologies effectively supported agricultural instruction by enhancing comprehension, participation, and practical application of concepts. The findings validate the effectiveness of technology-enhanced learning environments in promoting better educational outcomes within technical-vocational subjects such as agriculture.

Moreover, the study identified a significant positive relationship between the extent of application of digital technology tools and students' agricultural learning outcomes. This relationship confirms that greater utilization of digital technologies corresponded with improved student performance in agricultural education. The findings highlight the critical role of digital integration in fostering more effective teaching-learning processes and strengthening students' preparedness for technology-oriented agricultural practices and future vocational opportunities.

The study contributes to the growing body of knowledge emphasizing the educational value of digital technologies in technical and vocational education. It demonstrates that technology-enhanced agricultural instruction promotes engagement, digital literacy, experiential learning, and academic achievement among secondary school learners. The findings further support the need for continuous institutional investments in digital infrastructure, instructional innovation, teacher capacity-building, and learner digital competency development to sustain effective technology integration in agricultural education.

Based on the findings, schools and educational leaders may continue strengthening digital technology integration through sustained professional development programs, improved technological accessibility, and the provision of high-quality digital instructional materials. Teachers may likewise continue designing interactive and learner-centered digital learning activities that encourage participation, collaboration, and practical application of agricultural concepts. Future researchers may further investigate other contextual factors influencing digital technology utilization and agricultural learning outcomes across broader educational settings and learner populations.

References

Ahmed, R., & El-Sayed, M. (2023). Integrating drone technology in agricultural education: Enhancing technical competencies among secondary learners. International Journal of Agricultural Science and Technology, 12(4), 211–225.

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Educational Research and Practice (QJERP)
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Garcia, P., & Evans, D. (2023). Gamified learning environments and student participation in secondary education. Computers and Education Open, 4, 100118. https://doi.org/10.1016/j.caeo.2022.100118

Johnson, T., & Mitchell, R. (2021). Comparative agricultural competencies of urban and rural secondary school students. Journal of Agricultural Education Research, 62(4), 201–217.

Kim, H., & Lee, S. (2022). Digital learning tools and agricultural competencies among Korean secondary students. Asian Journal of Agricultural Education, 10(2), 132–149.

Lee, H., & Kim, J. (2022). Educational mobile applications and student usage patterns in South Korean secondary schools. Educational Media International, 59(1), 44–60. https://doi.org/10.1080/09523987.2022.2034567

Lee, S., & Park, M. (2021). Social media engagement and digital literacy development among university students. Interactive Learning Environments, 29(7), 1043–1057. https://doi.org/10.1080/10494820.2020.1711775

Martinez, A., & López, E. (2024). Teachers' digital competencies and information literacy integration in classroom instruction. International Journal of Educational Development, 102, 102890. https://doi.org/10.1016/j.ijedudev.2023.102890

Patel, R., & Singh, V. (2022). Accessibility, digital literacy, and student engagement in Indian higher education institutions. International Journal of E-Learning and Distance Education, 37(2), 95–113.

Santos, M., & Villanueva, R. (2024). Blended learning approaches in agricultural education among Filipino college students. Asia-Pacific Journal of Educational Technology, 20(1), 134–152.

Silva, J., & Pereira, C. (2023). Digital literacy and remote learning adaptation among Portuguese secondary school students. European Journal of Educational Research, 12(3), 1401–1415. https://doi.org/10.12973/eu-jer.12.3.1401

Smith, J., & Zhang, Y. (2021). Mobile learning applications and agricultural skill development among college students. Journal of Agricultural Education and Extension, 27(5), 561–578. https://doi.org/10.1080/1389224X.2021.1912321

Thompson, K., Rivera, J., & Brown, P. (2023). Student engagement and participation in virtual classroom environments. Computers in Human Behavior Reports, 10, 100282. https://doi.org/10.1016/j.chbr.2023.100282

Vygotsky, L. S. (1978). Mind in society: The development of higher psychological processes. Harvard University Press.

Zhang, Y., Li, X., & Chen, H. (2022). Information literacy instruction and critical evaluation skills among Chinese secondary school students. Journal of Information Literacy Education, 14(2), 121–139.

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Educational Research and Practice (QJERP)
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Ethical Approval Statement

This study was conducted in accordance with established ethical standards for educational research involving human participants. Prior to the conduct of the study, permission was secured from the appropriate school authorities of the participating secondary school in the Division of Cavite. Informed consent was obtained from all teacher participants, while parental consent and student assent were secured for minor student respondents. Participants were informed of the objectives, procedures, and voluntary nature of the study, including their right to withdraw participation at any stage without penalty. Confidentiality, anonymity, and privacy of all respondents were strictly maintained throughout the research process. All data gathered were used solely for academic purposes and were handled with utmost integrity, honesty, and transparency in accordance with ethical research principles.

Author Bio

Charisse H. Reyes is a professional teacher currently affiliated with Bulihan Integrated National High School, where she has served since 2008. She earned her Master of Arts in Education major in Technology and Livelihood Education from University of Perpetual Help System DALTA and previously completed her Bachelor in Business Teacher Education at Polytechnic University of the Philippines. Her professional experience includes serving as Curriculum Coordinator, Guidance Personnel Coordinator, and School-Based Feeding Program Coordinator. Her research interests focus on technology integration in education, agricultural learning, production, landscaping and garden design, and innovative instructional practices in Technology and Livelihood Education. She is committed to advancing learner-centered and technology-responsive educational approaches that strengthen practical competencies and academic achievement among secondary school learners.

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